

Digital Signal Processing: Mandatory Project

Analysis of the 15 January 2022 Hunga Tonga Eruption

Your Name / Candidate Number

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1 Task 1: The Overview

1.1 Task 1a

The provided dataset containing the HDF5 sensor time series was parsed. Figure 1 displays the geographic map of the event location and the respective recording stations.

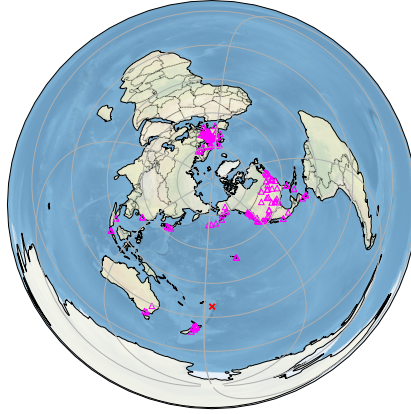


Figure 1: Geographic locations of the sensors and the Hunga Tonga volcano.

1.2 Task 1b

Based on the station metadata, the calculated great circle distances from the Hunga Tonga event are as follows:

- **Shortest distance:** 2137.71 km
- **Longest distance:** 17687.13 km

1.3 Task 1c

Figure 2 displays all sorted distances between the event location and the 201 station locations.

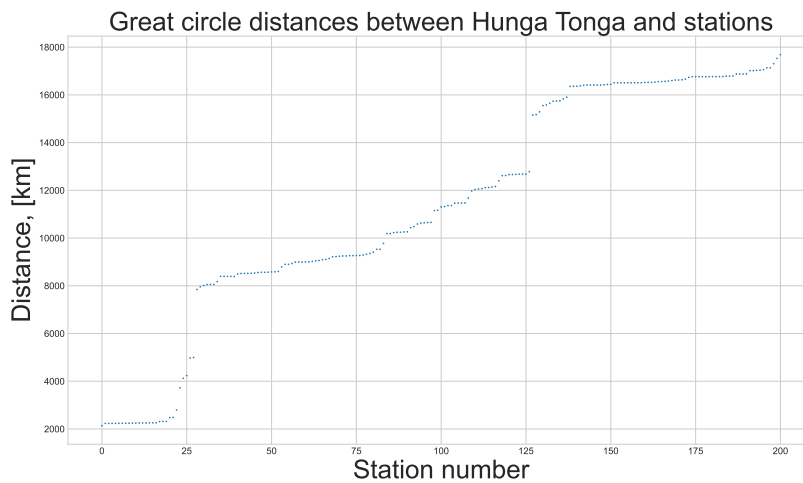


Figure 2: Sorted great circle distances from Hunga Tonga to all recording stations.

2 Task 2: Data Filtering

2.1 FIR Filter Impulse Responses (Task 2a)

The three provided FIR filter impulse responses, $h_1[n]$, $h_2[n]$, and $h_3[n]$, are plotted in Figure 3.

Figure 3: Impulse responses of the three FIR filters.

2.2 Custom Convolution Implementation (Task 2b)

To filter the input signal $x[n]$ using the FIR filter $h[n]$, the following convolution operation was implemented:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \quad (1)$$

The custom routine includes the `hlen_choice` parameter. A verification of this implementation against a toy signal (compared to the standard built-in function) is shown in Figure 4.

Figure 4: Verification of the custom convolution function using a toy signal.

2.3 Discrete-Time Fourier Transform Implementation (Task 2c)

The Discrete-Time Fourier Transform (DTFT) of the sequences was computed using the definition:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} \quad (2)$$

The implemented function takes the inputs `h`, `N`, and `fs`, and outputs both the complex vector $H(e^{j\omega})$ and the frequency axis in Hertz.

2.4 Frequency Spectra of Filters (Task 2d)

The absolute values of the frequency spectra, $|H_1(e^{j\omega})|$, $|H_2(e^{j\omega})|$, and $|H_3(e^{j\omega})|$, are plotted against the physical frequency F in Figure 5.

Figure 5: Absolute frequency spectra of the three provided FIR filters.

2.5 Filter Classification (Task 2e)

Based on the frequency spectra observed in Figure 5, the filters are classified as follows:

- $h_1[n]$ is a [Insert classification: lowpass/bandpass/highpass] filter.
- $h_2[n]$ is a [Insert classification: lowpass/bandpass/highpass] filter.
- $h_3[n]$ is a [Insert classification: lowpass/bandpass/highpass] filter.

2.6 Global Data Filtering (Task 2f)

All 201 input time series were successfully filtered using the three respective impulse responses via built-in convolution routines for execution speed.

2.7 Section Plot of Propagating Waves (Task 2g)

To recreate findings similar to Matoza et al. (2022), the data was bandpass filtered between 0.01 and 2 Hz. Filter [Insert Filter Name, e.g., $h_2[n]$] was chosen for this task because [Insert your justification here based on your DTFT analysis].

Figure 6 shows the resulting section plot with time on the x-axis and distance from Hunga Tonga on the y-axis.

Figure 6: Section plot of the filtered waveforms (0.01–2 Hz) demonstrating global wave propagation.

3 Task 3: Celerity Estimation for the Acoustic Wave

3.1 Arrival Time Picking (Task 3a)

A custom routine was developed to loop over all station signals. For each station, the raw and filtered traces were analyzed to manually pick the arrival time of the higher-frequency infrasound wave. The picked times were saved for subsequent analysis.

3.2 Celerity Estimation and Literature Comparison (Task 3b)

Figure 7 plots the estimated travel times against the event-to-station distances. Identified outliers (caused by noise or competing arrivals) are marked in [Insert Color, e.g., red] and were excluded from the linear regression.

Figure 7: Linear fit of travel time versus distance for celerity estimation. Outliers are excluded from the fit.

By fitting a straight line to the accepted distance/time data, the slope yielded an estimated celerity of [Insert Value] m/s.

Comparing this result to the literature, this estimate aligns with typical celerities of infrasound waves as reported by Vergoz et al. (2022) and Matoza et al. (2022), which state [Insert brief comparison/typical values from the papers].

Appendix: Code Links & Structure

As per the submission requirements, all code is properly commented and attached in the submission archive.

- **Main Run Script:** `main.py` / `main.m` (Running this script without inputs generates all figures included in this report).
- **Filtering Module:** `filters.py` / `filters.m` (Contains the custom convolution and DTFT functions).
- **Analysis Module:** `celerity.py` / `celerity.m` (Contains the arrival picking tool and linear fit functions).